

Simulation Study

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Loading libraries

```
library(sf)
```

Linking to GEOS 3.12.0, GDAL 3.9.1, PROJ 9.4.1; sf_use_s2() is TRUE

```
library(ggplot2)
library(spdep)
```

Loading required package: spData

To access larger datasets in this package, install the spDataLarge package with: ``install.packages('spDataLarge', repos='https://nowosad.github.io/drat/', type='source')``

```
library(Matrix)
library(MASS)
library(dplyr)
```

Attaching package: 'dplyr'

The following object is masked from 'package:MASS':

`select`

The following objects are masked from 'package:stats':

`filter, lag`

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(sp)
library(openxlsx)
library(adespatial)
```

Registered S3 methods overwritten by 'adegraphics':

method	from
biplot.dudi	ade4
kplot.foucart	ade4
kplot.mcoa	ade4
kplot.mfa	ade4
kplot.pta	ade4
kplot.sepan	ade4
kplot.statis	ade4
scatter.coa	ade4
scatter.dudi	ade4
scatter.nipals	ade4
scatter.pco	ade4
score.acm	ade4
score.mix	ade4
score.pca	ade4
screeplot.dudi	ade4

Registered S3 method overwritten by 'ape':

method	from
plot.mst	spdep

Registered S3 method overwritten by 'adespatial':

method	from
plot.multispati	adegraphics

```
library(igraph)
```

Attaching package: 'igraph'

The following objects are masked from 'package:dplyr':

as_data_frame, groups, union

The following objects are masked from 'package:stats':

decompose, spectrum

The following object is masked from 'package:base':

union

```
library(mclust)
```

Package 'mclust' version 6.1.2

Type 'citation("mclust")' for citing this R package in publications.

Attaching package: 'mclust'

The following object is masked from 'package:dplyr':

count

```
library(SpatialExperiment)
```

Loading required package: SingleCellExperiment

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'matrixStats'

The following object is masked from 'package:mclust':

count

The following object is masked from 'package:dplyr':

count

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: GenomicRanges

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:igraph':

components, union

The following object is masked from 'package:dplyr':

explain

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:igraph':

normalize, path

The following object is masked from 'package:dplyr':

combine

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

Filter, Find, Map, Position, Reduce, anyDuplicated, aperm, append,
as.data.frame, basename, cbind, colnames, dirname, do.call,
duplicated, eval, evalq, get, grep, grepl, is.unsorted, lapply,
mapply, match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
rank, rbind, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Loading required package: S4Vectors

Attaching package: 'S4Vectors'

The following objects are masked from 'package:dplyr':

first, rename

The following objects are masked from 'package:Matrix':

expand, unname

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

I, expand.grid, unname

Loading required package: IRanges

Attaching package: 'IRanges'

The following object is masked from 'package:sp':

%over%

The following objects are masked from 'package:dplyr':

collapse, desc, slice

Loading required package: Seqinfo

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

Found more than one class "Annotated" in cache; using the first, from namespace 'RNeXML'

Also defined by 'S4Vectors'

Found more than one class "Annotated" in cache; using the first, from namespace 'RNeXML'

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Also defined by 'S4Vectors'

Found more than one class "Annotated" in cache; using the first, from namespace 'RNeXML'

Also defined by 'S4Vectors'

```
library(Banksy)
library(scater)
```

Loading required package: scuttle

Data Simulation

- Generating 'n' coordinates inside dim x dim grid

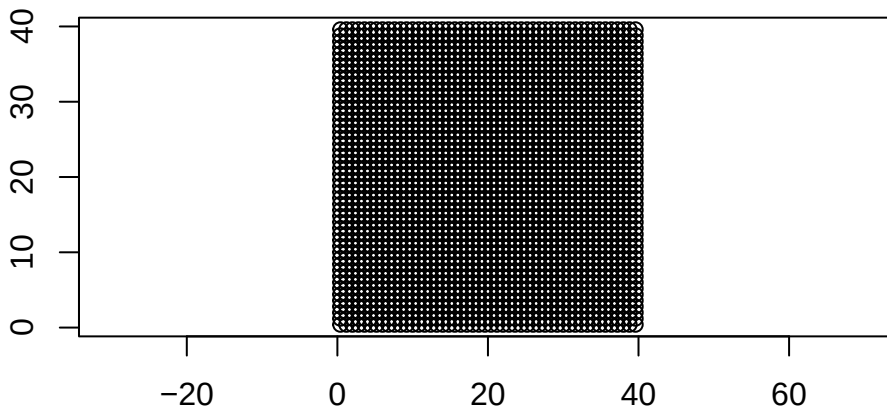
```
# Grid dimension
dim = 40

# Number of points to simulate
n = 2500
nrw = 50
ncl = 50

# Defining the domain size
sfc = st_sfc(
  st_polygon(list(rbind(c(0,0), c(dim,0), c(dim,dim), c(0,dim), c(0,0))))
)

# Simulating the coordinates
coordinates = st_make_grid(sfc, n=c(nrw,ncl), what = "centers")

plot(coordinates, axes = TRUE)
```



- Capturing different regions inside the polygon

```
# Defining the circle 1
center1 <- c(33,8)
radius1 <- 5
circle1 <- st_buffer(st_sfc(st_point(center1)), dist = radius1)

# Defining the circle 2
```



```

center2 <- c(8,33)
radius2 <- 5
circle2 <- st_buffer(st_sfc(st_point(center2)), dist = radius2)

# Defining the quadrangle
length_quad <- 9
corner_coordinates_1 <- list(
  rbind(
    c(3,3),
    c(3+length_quad,3),
    c(3+length_quad,3+length_quad),
    c(3,3+length_quad), c(3,3))
  )

square <- st_polygon(corner_coordinates_1)

# Defining the triangle
corner_coordinates_2 <- list(rbind(c(28,31), c(38,31), c(33,38), c(28,31)))
triangle <- st_polygon(corner_coordinates_2)

# Defining center larger circle and smaller circle at the center
centers <- c(20,20)
radius_l <- 8
radius_s <- 5

large_circle <- st_buffer(st_sfc(st_point(centers)), dist = radius_l)
small_circle <- st_buffer(st_sfc(st_point(centers)), dist = radius_s)

# Create a ring by subtracting the smaller circle from the larger circle
ring <- st_difference(large_circle, small_circle)

```

- Capturing the coordinates inside different regions

```

# Circle 1
circle_1_coords <- st_intersection(coordinates, circle1)
# Quadrangle
quad_coords <- st_intersection(coordinates, square)
# Circle 2
circle_2_coords <- st_intersection(coordinates, circle2)
# Triangle
tri_coords <- st_intersection(coordinates, triangle)
# Ring

```

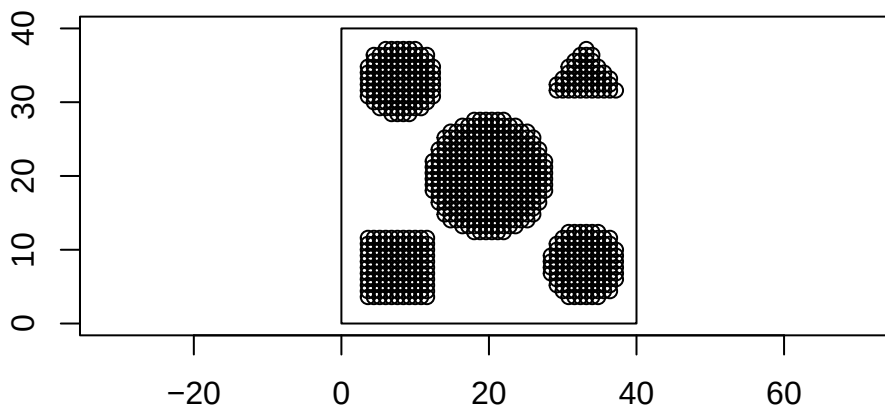
```
ring_coords <- st_intersection(coordinates, ring)
# Circle inside the ring
circle_s_coords <- st_intersection(st_intersection(coordinates, small_circle))
```

- Capturing the number of coordinates inside each region

```
num_circle_1 <- length(circle_1_coords)
num_quad <- length(quad_coords)
num_circle_2 <- length(circle_2_coords)
num_tri <- length(tri_coords)
num_ring <- length(ring_coords)
num_circle_s <- length(circle_s_coords)
```

- Plotting all the captured regions

```
cap_coords <- st_sf(
  coords = c(
    circle_1_coords,
    circle_2_coords,
    quad_coords,
    tri_coords,
    circle_s_coords,
    ring_coords
  )
)
plot(sfc, axes = TRUE)
plot(cap_coords, add = TRUE)
```



- Coordinates in the un-captured region (noise)

```

# Function to create an "sf object" of the union of the shapes
union_sf <- function(sf_objects) {
  result <- sf_objects[[1]]
  for (i in 2:length(sf_objects)) {
    result <- st_union(result, sf_objects[[i]])
  }
  return(result)
}

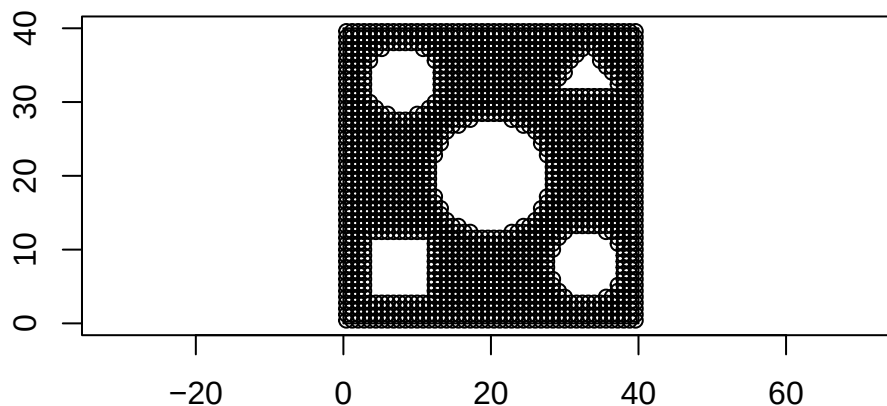
sf_list <- list(circle1, circle2, square, triangle, large_circle)

union_result <- union_sf(sf_list)

# Coordinates of the noise
non_coords <- st_sf(
  coords = st_difference(coordinates, union_result)
)

plot(sfc, axes = TRUE)
plot(non_coords, add = TRUE)

```



- Function for simulations from spatial models

CAR

```

# Function to simulate for the selected regions
simulate_CAR <- function(n, coords, k, mu, tau_sq, eta){

```

```

# Creating k-nearest neighbors
nb_simulation <- knn2nb(knearneigh(coords, k))

# Making the neighborhood structure un-directional
nb_1_sym <- make.sym.nb(nb_simulation)

# Weight matrix for simulation
w_mat <- nb2mat(nb_1_sym, style = "W")

# Defining C matrix
c <- eta*w_mat

# Defining mean matrix
mu_mat <- rep(mu,n)

# Defining sigma matrix
# Defining sigma matrix
identity <- diag(seq(1,1,length=n))
sigma_mat <- tau_sq*solve(identity-c)

# Simulating from joint distribution of the variables from CAR model (MVN)
z_CAR <- mvrnorm(n=1, mu_mat, sigma_mat)
z <- as.vector(z_CAR)

return(z)
}

```

- Function for simulating noise

```

simulate_noise <- function(n, coords, p, mu, sigma){

# Initiating mean and sigma
mu_mat <- rep(mu,p)

sigma_mat <- diag(rep(sigma, p))

# Simulating data from the multivariate normal distribution
noise <- mvrnorm(n=n,mu=mu_mat, Sigma = sigma_mat)
colnames(noise) <- paste0("Z",1:p)
data <- cbind(coords, noise)

```

```

return(data)
}

```

- Function for calculating MMD^2

```

compute_mmd_sq <- function(
  sample1_idx,
  sample2_idx,
  dist_matrix
) {

  N <- length(sample1_idx)
  M <- length(sample2_idx)

  # First term:  $1/N^2 \sum K(x_n, x_n')$ 
  term1 <- sum(1/(sqrt(1+(dist_matrix[sample1_idx, sample1_idx])^2))) / (N * N)

  # Second term:  $-2/NM \sum K(x_n, x_m')$ 
  term2 <- -2 * sum(1/(sqrt(1+(dist_matrix[sample1_idx, sample2_idx])^2))) / (N * M)

  # Third term:  $1/M^2 \sum K(x'_m, x'_m')$ 
  term3 <- sum(1/(sqrt(1+(dist_matrix[sample2_idx, sample2_idx])^2))) / (M * M)

  #  $\text{MMD}^2$  value
  mmd_sq <- term1 + term2 + term3
  return(mmd_sq)
}

```

- Function for creating blocks based on features inside each region (Using KMeans)

```

create_blocks <- function(data_regions, k_sim, p) {

  # Initialize an empty list to blocks of each region
  blk_data <- list()
  regions <- unique(data_regions$region)

  # For each region perform kmeans blocking
  for (i in 1:length(regions)) {

    region_data <- data_regions[data_regions$region == regions[i],]
    num_blks <- floor(nrow(region_data)/k_sim)
    # If the cluster is smaller than the neighborhood size, use 1 bloc

```

```

num_blks <- ifelse(num_blks == 0, 1, num_blks)

df <- region_data[,2:(p+1)] |>
  st_drop_geometry()

kmeans_result <- kmeans(df, num_blks)$cluster

region_data <- cbind(region_data, polygon_id = kmeans_result)

blk_data[[i]] <- region_data
}

combined_sf_df <- do.call(rbind, blk_data)

return(combined_sf_df)
}

```

- Function to perform block permutation

```

compute_perm_mmd2 <- function(cl1,
                              cl2,
                              dist_matrix,
                              df){

  # Saving the original order
  df$point <- 1:nrow(df)

  # Extracting rows such that observation of the smaller cluster is at top
  cluster_list <- list(cl1,cl2)
  sorted_list <- cluster_list[order(sapply(cluster_list, length))]

  df <- df[c(sorted_list[[1]], sorted_list[[2]]),]

  df <- df |>
    group_by(region, polygon_id) |>
    mutate(block_id = cur_group_id()) |>
    ungroup()

  # Saving the number of observations in of the smallest cluster
  n1 <- sum(df$region == unique(df$region)[1])

```

```

blk_id1 <- c()
n1_permuted <- 0

while (n1_permuted < n1) {
  new_block <- sample(
    setdiff(
      unique(df$block_id),
      blk_id1),
    1
  )
  blk_id1 <- c(blk_id1, new_block)

  # The number of data points in the sampled blocks
  n1_permuted <- length(df$polygon_id[df$block_id %in% blk_id1])
}

df <- df %>%
  mutate(
    permuted_region = ifelse(
      block_id %in% blk_id1,
      unique(df$region)[1],
      unique(df$region)[2]
    )
  )

perm_mmd_sq <- compute_mmd_sq(
  df$point[df$permuted_region == unique(df$permuted_region)[1]],
  df$point[df$permuted_region == unique(df$permuted_region)[2]],
  dist_matrix
)
return(perm_mmd_sq)
}

```

- Function to conduct pairwise MMD test

```

MMD_pairwise_test <- function(data_regions, dist_matrix, nperm) {

  # list to store pairwise results
  pairwise_results <- list()

  # Unique cluster pairs
  cluster_pairs <- combn(unique(data_regions$region), 2)
}

```

```

for (i in 1:ncol(cluster_pairs)) {

  cluster_1 <- which(data_regions$region == cluster_pairs[1, i])
  cluster_2 <- which(data_regions$region == cluster_pairs[2, i])

  obs_mmd_sq <- compute_mmd_sq(cluster_1, cluster_2, dist_matrix)

  null_mmd_sq <- lapply(
    1:nperm,
    function(i){
      compute_perm_mmd2(cluster_1, cluster_2, dist_matrix, data_regions)
    }
  ) |> unlist()

  p_value <- mean(null_mmd_sq >= obs_mmd_sq)
  pairwise_results[[i]] <- c(cluster_pairs[1, i], cluster_pairs[2, i], p_value)
}

# combine list elements to a datarame
results_df <- as.data.frame(do.call(rbind, pairwise_results))
colnames(results_df) <- c("region_1", "region_2", "p_value")

# Get the adjusted p-value
results_df$adj_p <- p.adjust(results_df$p_value, method = "BH")

return(results_df)
}

```

Running simulations

```

runs <- 100 # Number of simulation runs
nperm <- 100 # Number of permutations
p_signal <- 5 # original number of signal variables
p_extra <- 3 # two added noise variables
p_total <- p_signal + p_extra
#p <- 5 # Number of variables
k_sim <- 8 # Number of neighbors for simulation

# Model parameters for Square, Circle 1 and Ring

```



```

mu_1 <- 4
tau_sq_1 <- 1
eta_1 <- 0.8

# Model parameters for Square, Circle 1 and Ring
mu_2 <- 10
tau_sq_2 <- 1
eta_2 <- 0.8

# Model parameters for noise
mu_noise <- 0
sigma_noise <- 1

# List of different regions in region set 1
regions_1 <- list(
  st_sf(coords = circle_1_coords),
  st_sf(coords = quad_coords),
  st_sf(coords = ring_coords)
)

# Number of coordinates for each region in set 1
regions_1_num <- c(
  num_circle_1,
  num_quad,
  num_ring
)

# List of different regions in region set 2
regions_2 <- list(
  st_sf(coords = circle_2_coords),
  st_sf(coords = tri_coords),
  st_sf(coords = circle_s_coords)
)

# Number of coordinates for each shape in set 2
regions_2_num <- c(
  num_circle_2,
  num_tri,
  num_circle_s
)

# Number of coordinates for noise

```

```

num_noise <- n - sum(regions_1_num, regions_2_num)

# Coordinates for simulating noise
noise_coords <- non_coords

# Create a workbook to save results
wb <- createWorkbook()

# List to store run, accuracy and ARI
result_eval <- list()

for (run in 1:runs) {

  # Initialize empty lists to store the results
  results_list_1 <- list()
  results_list_2 <- list()

  for(i in 1:length(regions_1_num)) {

    # Simulating data for the regions in set 1
    sim_data_1 <- data.frame(
      lapply(
        1:p_signal,
        function(x) simulate_CAR(
          regions_1_num[i],
          regions_1[[i]],
          k_sim, mu_1, tau_sq_1, eta_1
        )
      ),
      lapply(
        1:p_extra,
        function(x) rnorm(regions_1_num[i], mean = 0, sd = 1)
      ),
      coords = regions_1[[i]]
    )

    names(sim_data_1)[1:p_total] <- c(paste0("Z", 1:p_total))
    results_list_1[[i]] <- sim_data_1

    # Simulating data for the regions in set 2

```

```

sim_data_2 <- data.frame(
  lapply(
    1:p_signal,
    function(x) simulate_CAR(
      regions_2_num[i],
      regions_2[[i]],
      k_sim, mu_2, tau_sq_2, eta_2
    )
  ),
  lapply(
    1:p_extra,
    function(x) rnorm(regions_2_num[i], mean = 0, sd = 1)
  ),
  coords = regions_2[[i]]
)

names(sim_data_2)[1:p_total] <- c(paste0("Z", 1:p_total))
results_list_2[[i]] <- sim_data_2
}

# Combining the two lists
results_list <- c(results_list_1, results_list_2)

# Name the result_list in the order of the simulated shapes
names(results_list) <- c(rep("clust1",3), rep("clust2",3))

# Convert the list to a data frame with a column for the name of the region
data_regions <- results_list |>
  bind_rows(.id = "sim_region")

# Simulate data for the noise coordinates and adding a region column
data_noise <- simulate_noise(num_noise, noise_coors, p_total, mu_noise, sigma_noise)
data_noise <- cbind(sim_region = "noise", data_noise)

# Convert data frame to an as sf object
data_regions<- st_as_sf(data_regions, sf_column_name = "coords")

# Merge noise and the clusters
data_regions <- rbind(data_regions,data_noise)

```

```

# Create a spatial experiment object to apply Banksy
assay <- data_regions[,-1] %>% st_drop_geometry() %>% t()
locs <- st_coordinates(data_regions)
se <- SpatialExperiment(assay = list(counts = assay), spatialCoords = locs)

# Compute mean gene expression values and AGFs
se <- computeBanksy(se, assay_name = 'counts', compute_agf = TRUE, k_geom = k_sim)

# Cluster using kmeans with 3 cluster centers
se <- clusterBanksy(
  se, use_agf = TRUE, use_pcs = FALSE, lambda = 0.8,
  algo = 'kmeans', kmeans.centers = 3
)

# Distance matrix on features
feature_distance <- data_regions[,2:(p_total+1)] |>
  st_drop_geometry() |>
  dist() |>
  as.matrix()

# Apply constrained clustering
nb <- knn2nb(knearneigh(
  as.data.frame(locs), k = k_sim)) %>%
  nb2listw(style = "W")

neighbors <- listw2sn(nb)[,1:2]

constr.clust <- constr.hclust(
  d = as.dist(feature_distance), method = "ward.D2",
  links = neighbors, coords = locs
)

# Optimal number of clusters for constrained is 7
data_regions$region <- cutree(constr.clust, 7)

# Creating blocks using k-means
data_regions <- create_blocks(data_regions, k_sim, p= p_total)

test_results <- MMD_pairwise_test(data_regions, feature_distance, nperm)

# Identify cluster pairs which didn't reject null
similar_pairs <- test_results[test_results$p_value >= 0.05, c("region_1", "region_2")]

```

```

# Convert the results to a graph
g <- graph_from_data_frame(similar_pairs, directed = FALSE)

# Obtain fully connected sub-graphs with a minimum of 3 nodes
cliques <- cliques(g, min = 3)
clique_nodes <- lapply(cliques, names)

# Re-partitioning the clusters
if (length(clique_nodes) != 0) {

  for (iter in 1:length(clique_nodes)) {
    data_regions$region <- replace(
      data_regions$region,
      data_regions$region %in% as.numeric(clique_nodes[[iter]]),
      letters[iter]
    )
  }

}

# Creating a copy to be saved in worksheet
test_results_cp <- test_results

# Comparing pairwise test accuracy (out of all pairwise test, how many gave the desired res
gt_test <- c(1,1,0,0,0,0,1,0,0,0,0,0,0,0,1,1,0,1,0,0)
rounded_p <- ifelse(test_results$adj_p >= 0.05, 1, 0)

# Classification table
table <- table(Actual = gt_test, Predicted = rounded_p)

# Save true positive, true negative, false positive and false negative
tp <- table[2,2]
tn <- table[1,1]
fp <- table[1,2]
fn <- table[2,1]

precision <- ifelse((tp + fp) == 0, 0, tp / (tp + fp))
recall <- ifelse((tp + fn) == 0, 0, tp / (tp + fn))

f1_score <- ifelse((precision + recall) == 0, 0, 2 * (precision * recall) / (precision + r

ari_repSpat <- adjustedRandIndex(data_regions$sim_region, data_regions$region)

```

```

ari_Banksy <- adjustedRandIndex(data_regions$sim_region, se$clust_M1_lam0.8_kmeans3)

result_eval[[run]] <- c(run,ari_repSpat,ari_Banksy,fn,fp,precision,recall,f1_score)

sheet_name <- paste("Run", run)
print(run)

# Add the data frame as a new sheet in the workbook
addWorksheet(wb, sheet_name)
writeData(wb, sheet = sheet_name, test_results_cp)
}

```

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 1

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 2

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 3

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 4

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 5

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 6

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 7

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 8

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 9

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 10

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

```
[1] 11
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

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Parameters: k_geom=8
```

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Done
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

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Parameters: k_geom=8
```

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Done
```

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Done
```

```
Centering
```

```
Done
```

```
[1] 12
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

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Parameters: k_geom=8
```

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Done
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 13

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 14

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 15

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 16

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 17

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 18

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 19

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 20

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 21

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 22

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 23

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 24

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 25

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 26

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 27

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 28

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 29

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 30

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 31

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 32

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 33

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

Warning: did not converge in 10 iterations

[1] 34

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 35

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 36

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 37

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 38

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 39

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 40

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 41

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

```
[1] 42
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

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Parameters: k_geom=8
```

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Done
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Computing neighbors...
```

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Spatial mode is kNN_median
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Parameters: k_geom=8
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Done
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Done
```

```
Centering
```

```
Done
```

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[1] 43
```

```
Computing neighbors...
```

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Spatial mode is kNN_median
```

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Parameters: k_geom=8
```

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Done
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
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Parameters: k_geom=8

Done

Done

Centering

Done

[1] 44

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 45

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 46

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 47

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 48

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 49

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 50

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 51

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 52

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 53

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 54

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 55

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 56

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 57

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 58

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 59

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

```
[1] 60
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

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Parameters: k_geom=8
```

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Done
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

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Parameters: k_geom=8
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Done
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Done
```

```
Centering
```

```
Done
```

```
[1] 61
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

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Parameters: k_geom=8
```

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Done
```

```
Computing neighbors...
```

```
Spatial mode is kNN_median
```

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 62

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 63

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 64

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 65

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 66

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 67

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 68

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 69

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 70

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 71

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 72

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 73

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 74

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 75

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 76

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 77

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 78

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 79

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 80

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 81

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 82

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 83

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 84

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 85

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 86

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 87

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 88

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 89

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 90

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 91

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 92

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 93

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 94

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 95

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 96

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 97

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 98

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 99

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Computing neighbors...

Spatial mode is kNN_median

Parameters: k_geom=8

Done

Done

Centering

Done

[1] 100

```
run_summary <- do.call(rbind, result_eval)
colnames(run_summary) <- c(
  "Run",
  "ARI_repSpat",
  "ARI_Banksy",
  "FN",
  "FP",
  "Precision",
  "Recall",
  "F1Score"
)
write.csv(run_summary, 'accuracies_n2500_p5_n3_rho0.8.csv', row.names = FALSE)
saveWorkbook(wb, "output_n2500_p5_n3_rho0.8.xlsx", overwrite = TRUE)
```

Save the results